

Introduction

Exercises ISLR – Ch.7

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Conceptual

Exercises: 2

Ex.2) Smoothing Splines

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$$

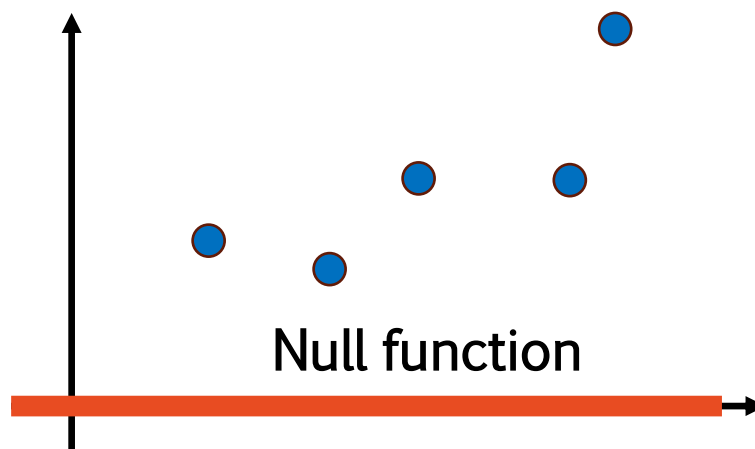
loss function
(data adjustment quality)

penalty term
(penalize changes, making
the function “smoother”)

Ex.2a) Smoothing Splines

$$\hat{g} = \arg \min_g \left(\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int [g^{(m)}(x)]^2 dx \right),$$

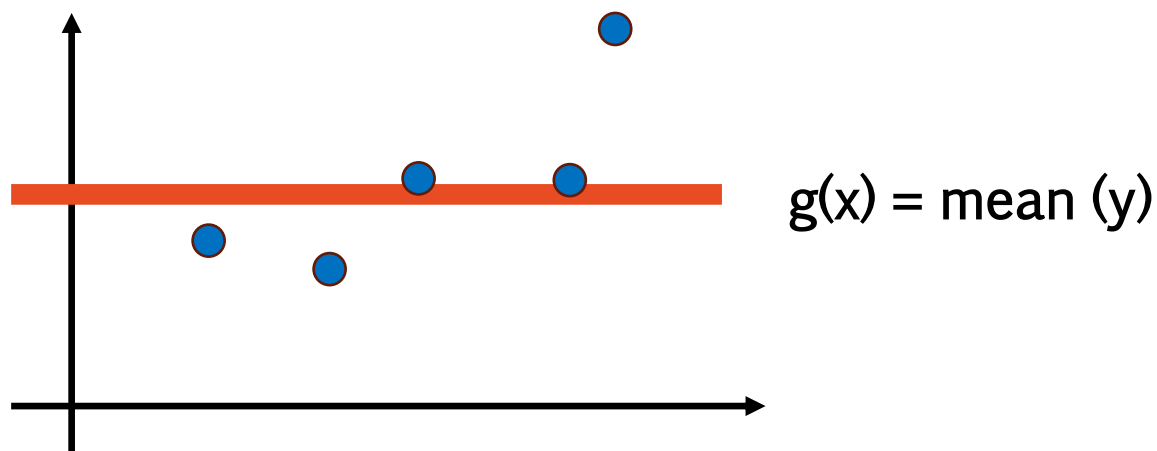
(a) $\lambda = \infty, m = 0 \rightarrow$ Extreme penalty to integral of $g(x) \rightarrow g(x) = 0$



Ex.2b) Smoothing Splines

$$\hat{g} = \arg \min_g \left(\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int [g^{(m)}(x)]^2 dx \right),$$

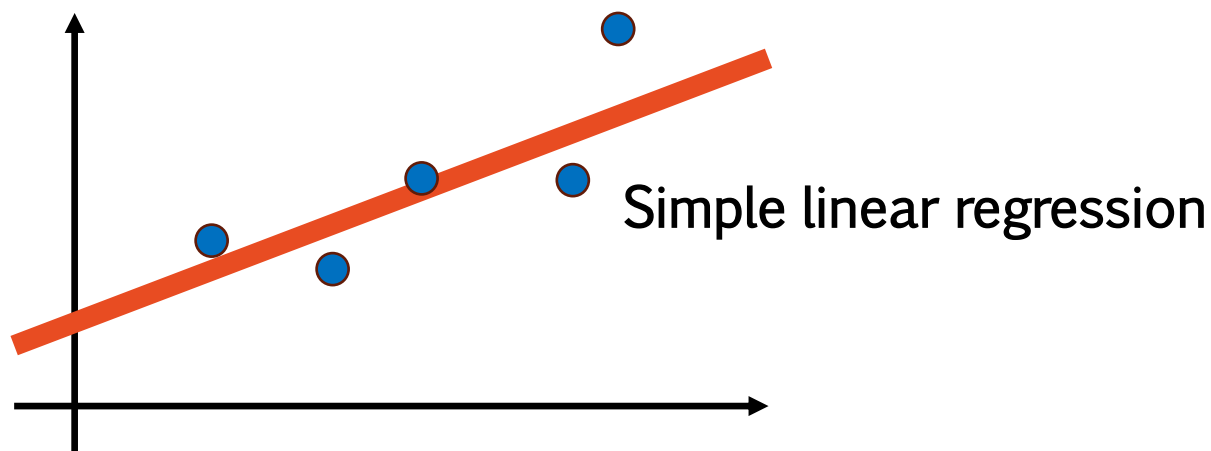
(b) $\lambda = \infty, m = 1 \rightarrow$ Extreme penalty to integral of $g'(x) \rightarrow g'(x) = 0 \rightarrow g(x) = k$



Ex.2c) Smoothing Splines

$$\hat{g} = \arg \min_g \left(\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int [g^{(m)}(x)]^2 dx \right),$$

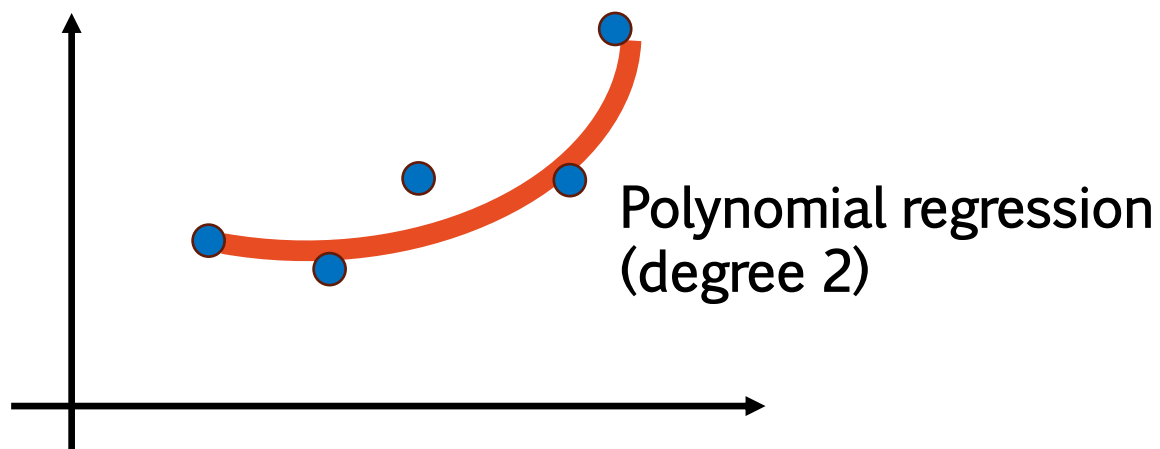
(c) $\lambda = \infty, m = 2 \rightarrow$ Extreme penalty to integral of $g''(x) \rightarrow g''(x) = 0 \rightarrow g'(x) = a$
 $\rightarrow g(x) = ax + b$



Ex.2d) Smoothing Splines

$$\hat{g} = \arg \min_g \left(\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int [g^{(m)}(x)]^2 dx \right),$$

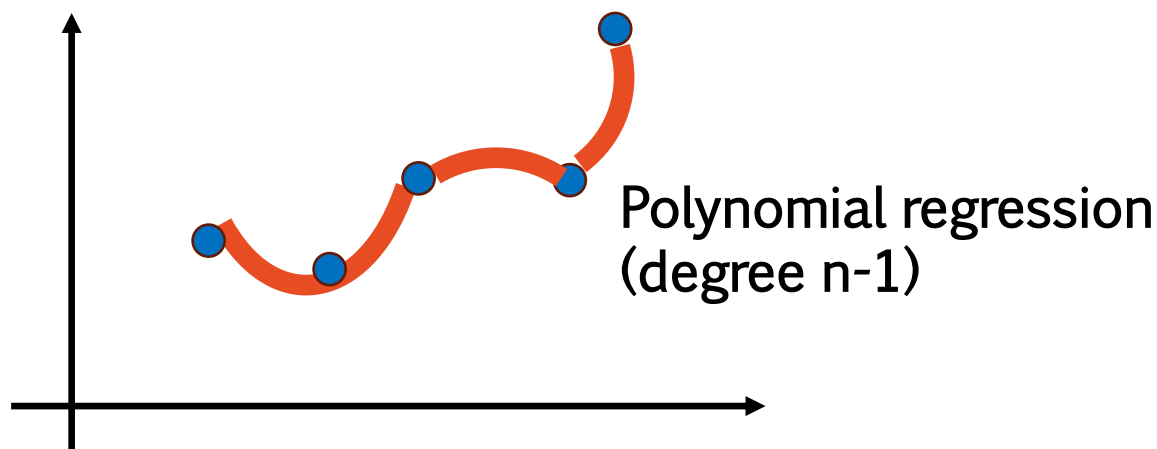
(d) $\lambda = \infty, m = 3 \rightarrow$ Extreme penalty to integral of $g'''(x) \rightarrow g'''(x) = 0 \rightarrow$
 $\rightarrow g''(x) = a \rightarrow g'(x) = ax + b \rightarrow g(x) = ax^2 + bx + c$



Ex.2e) Smoothing Splines

$$\hat{g} = \arg \min_g \left(\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int [g^{(m)}(x)]^2 dx \right),$$

(e) $\lambda = 0, m = 3 \rightarrow$ No penalty to derivatives $\rightarrow$ minimize loss function $\rightarrow$ polynomial passing through all points (n-1 order)





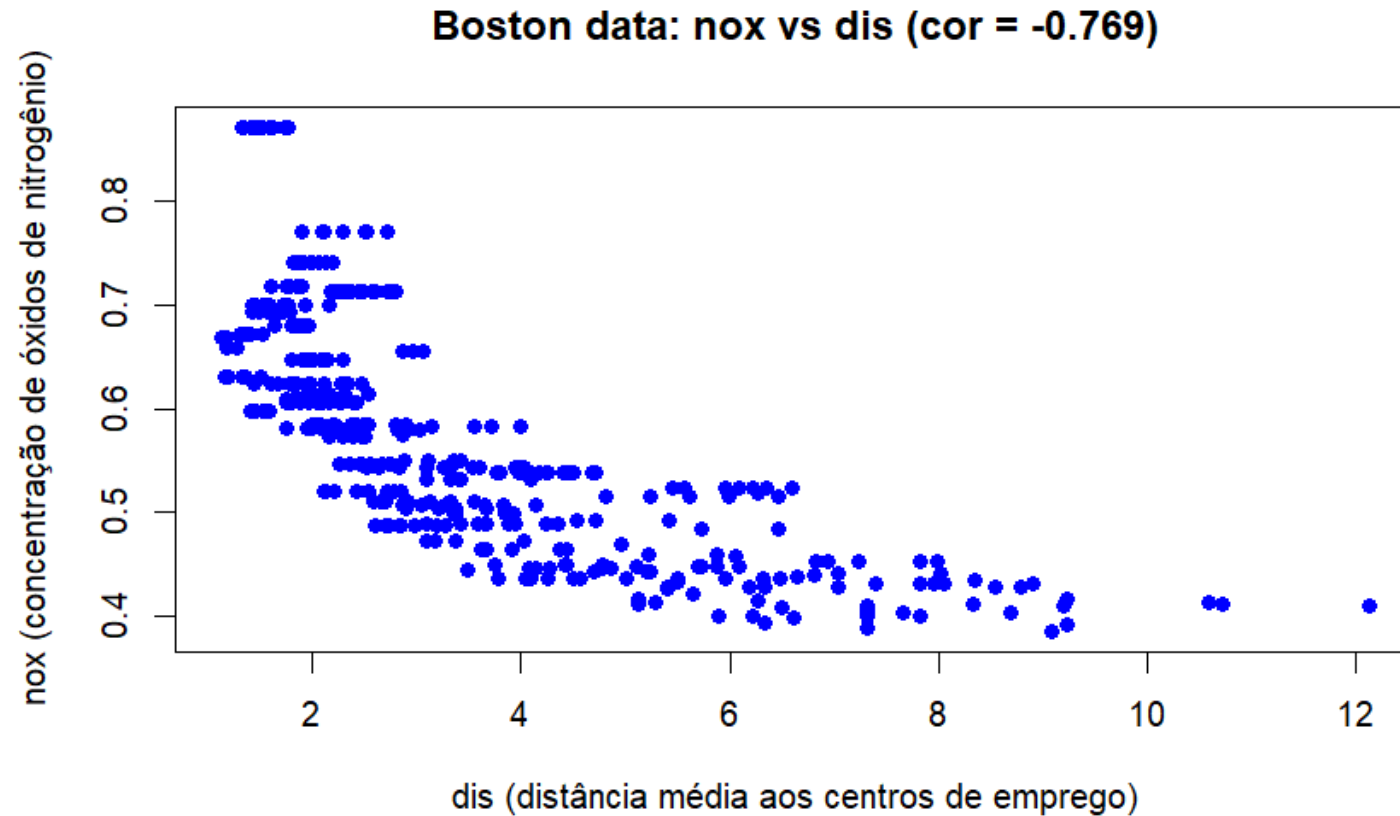
Applied

Exercise 9 – Boston dataset

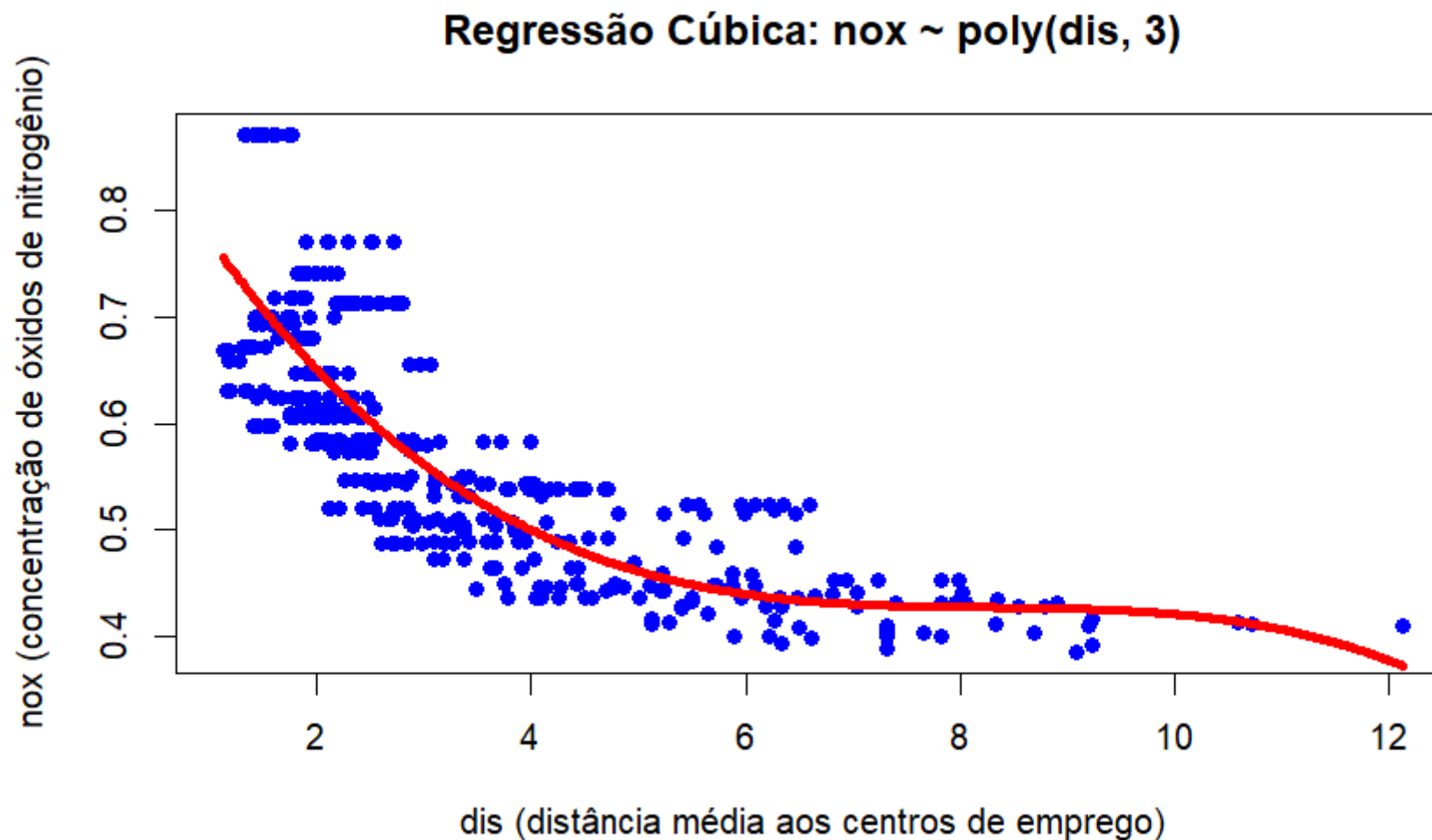
Ex.9) Regressions: $\text{Nox} = f(\text{Dis})$



Regressions: Predict nitrogen oxide levels as a function of distance to employment centers



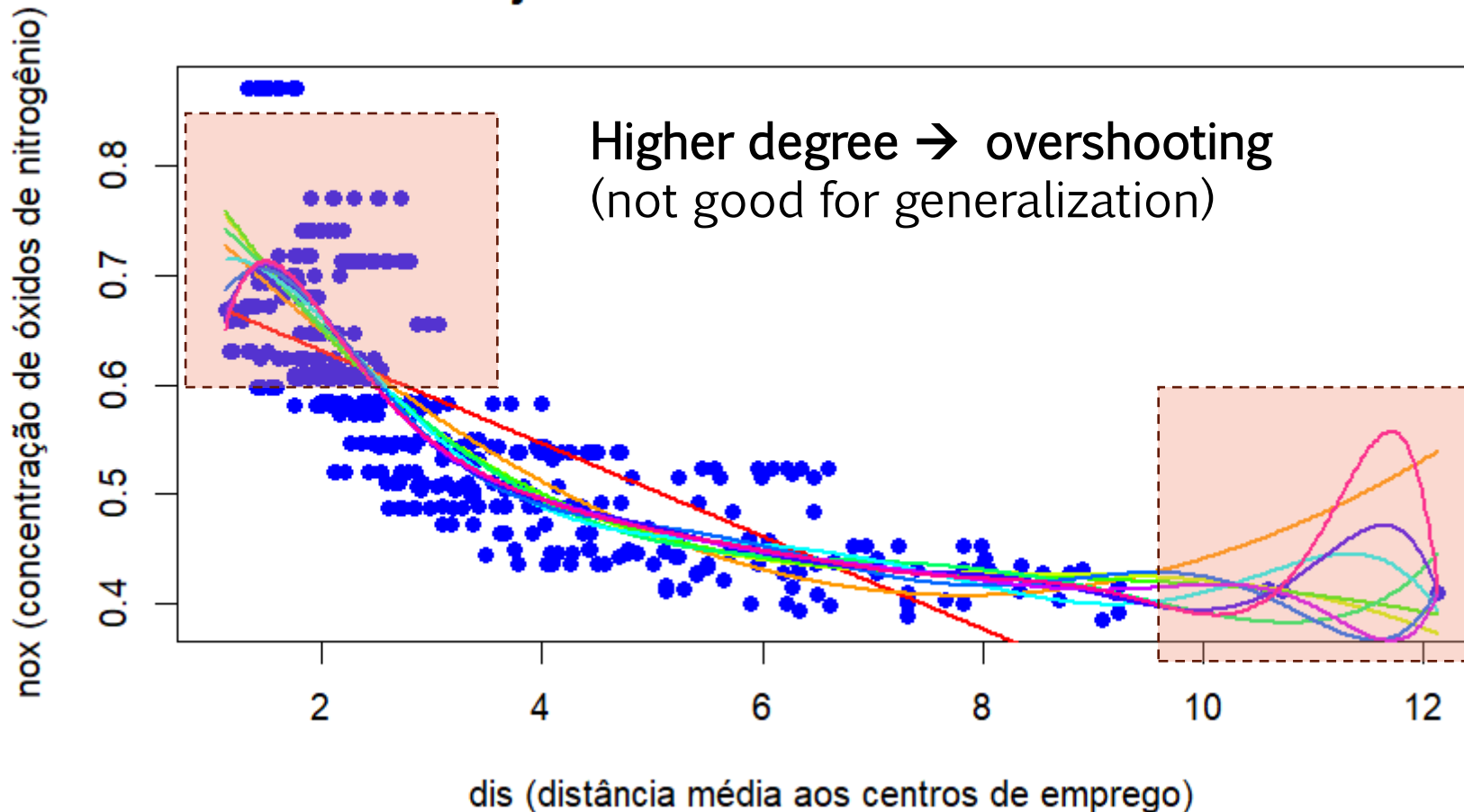
Ex.9a) Polynomial Regression: Cubic



Ex.9b) Polynomial Regression: 1 to 10



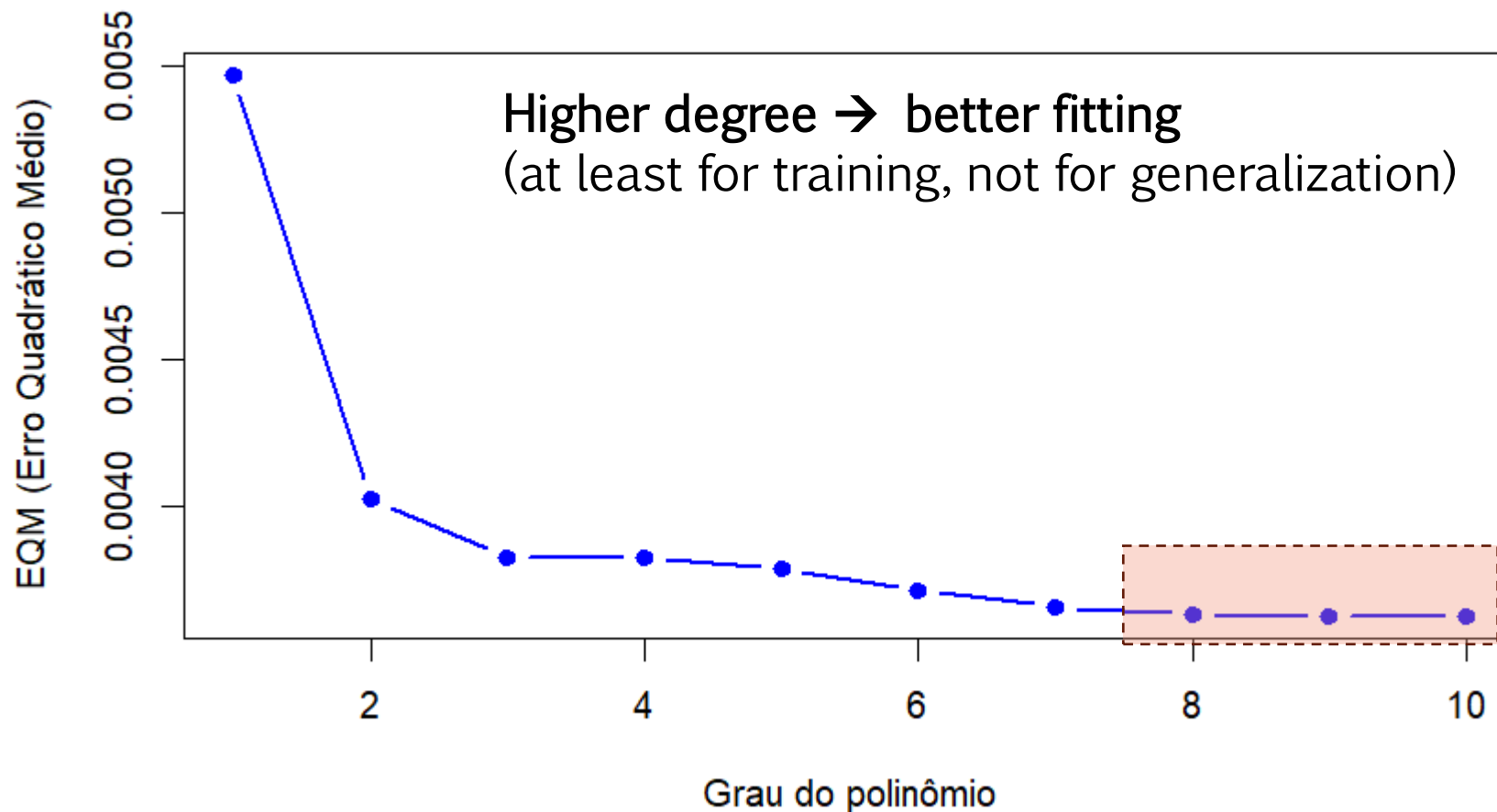
Ajustes Polinomiais: Graus 1 a 10



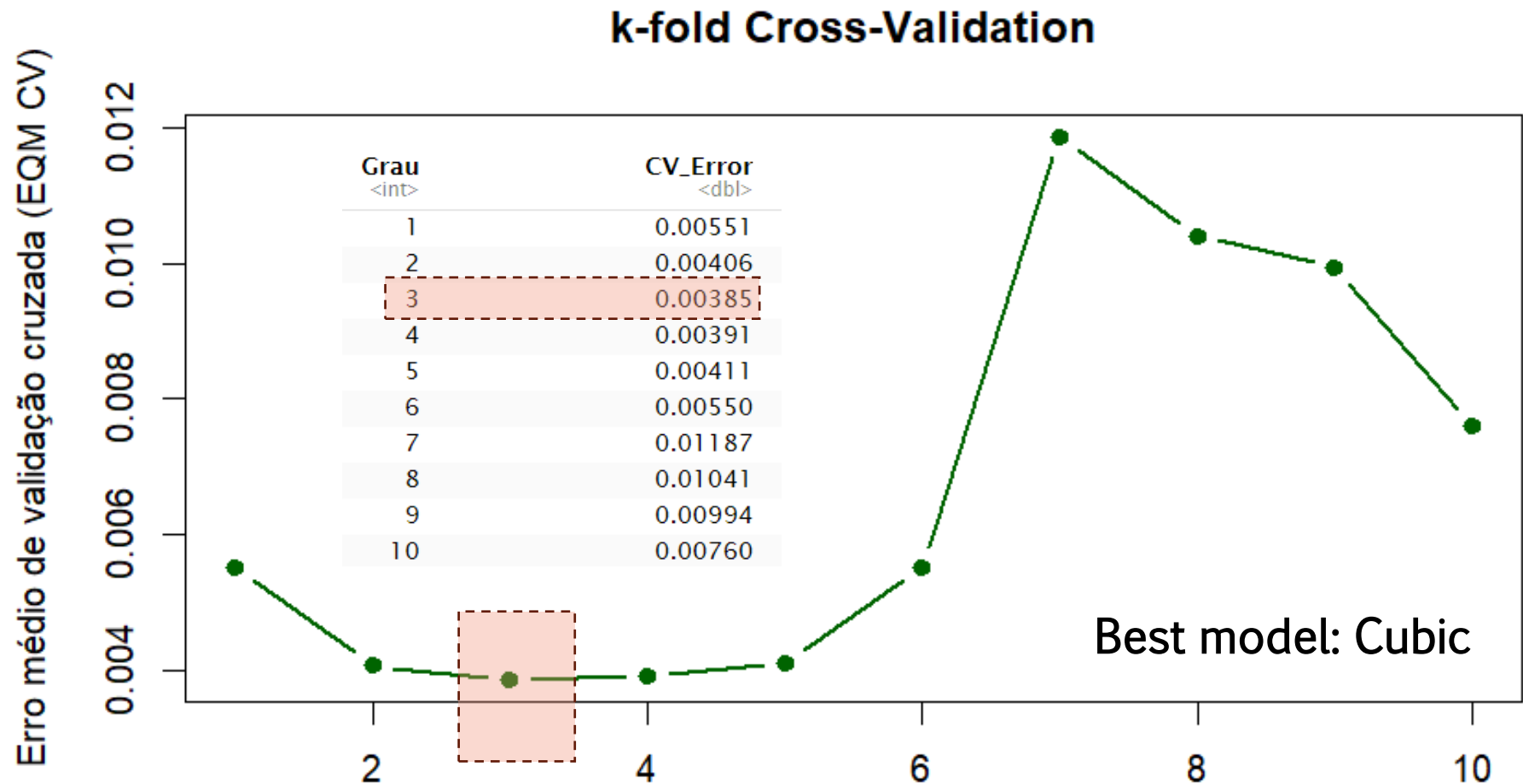
Ex.9b) Polynomial Regression: 1 to 10



EQM por Grau do Polinômio

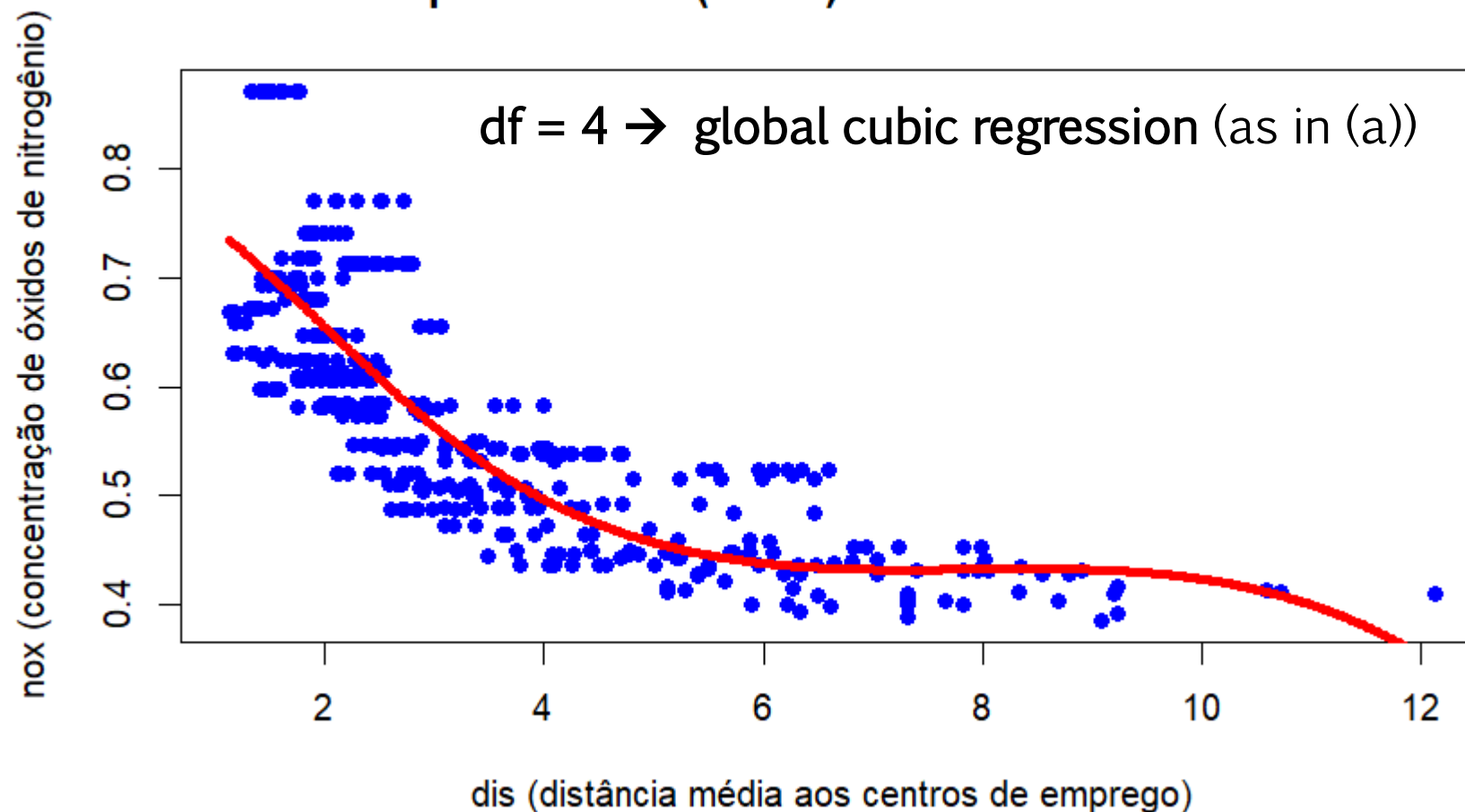


Ex.9c) Polynomial Regression: 10-fold CV



Ex.9d) Cubic Splines: $df = 4 \rightarrow \text{knots} = 0$

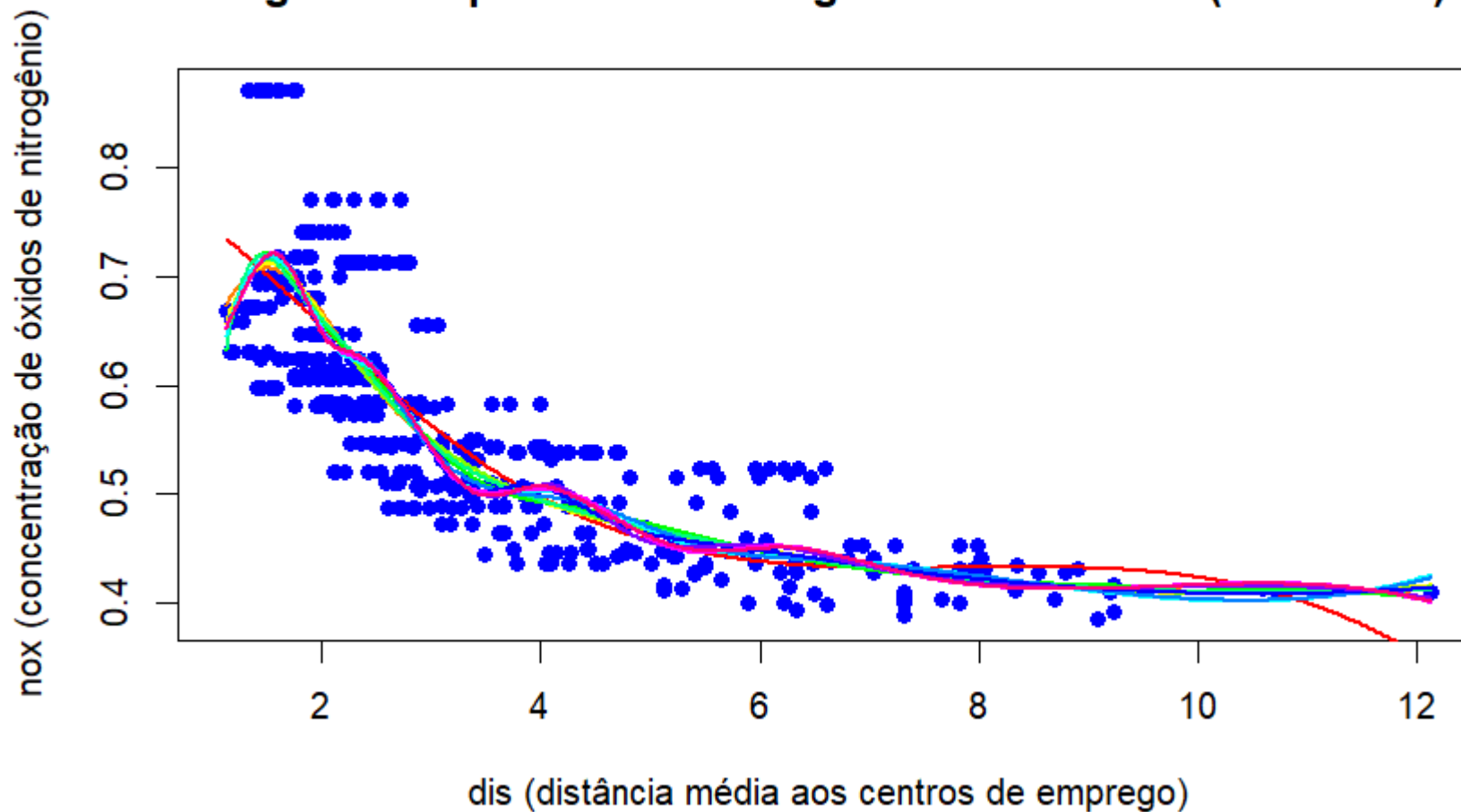
Spline cúbico ($df = 4$) — sem nós internos



Ex.9e) Cubic Splines: df = 4 to 15



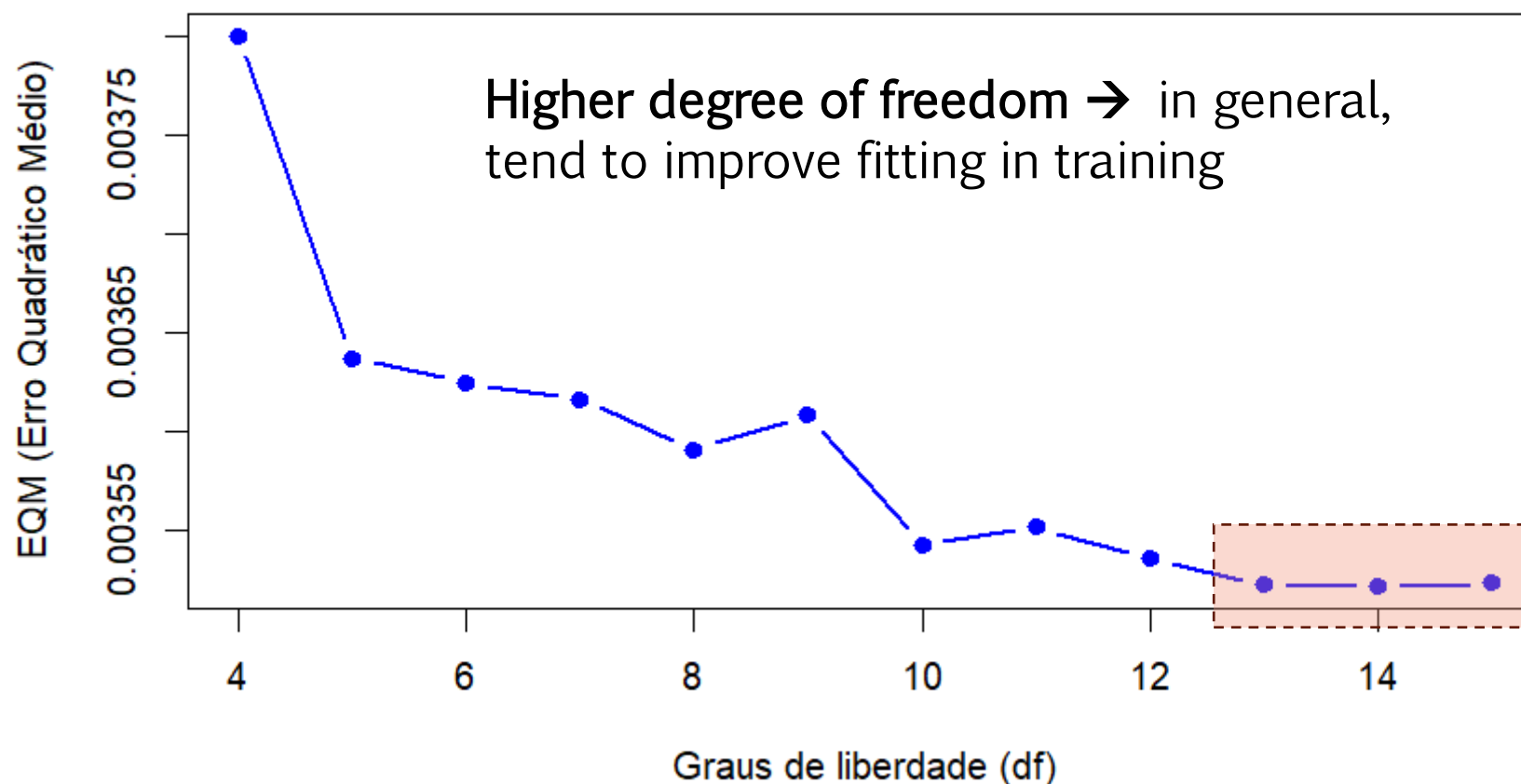
Regressão Spline: diferentes graus de liberdade (df = 4 a 15)



Ex.9e) Cubic Splines: df = 4 to 15



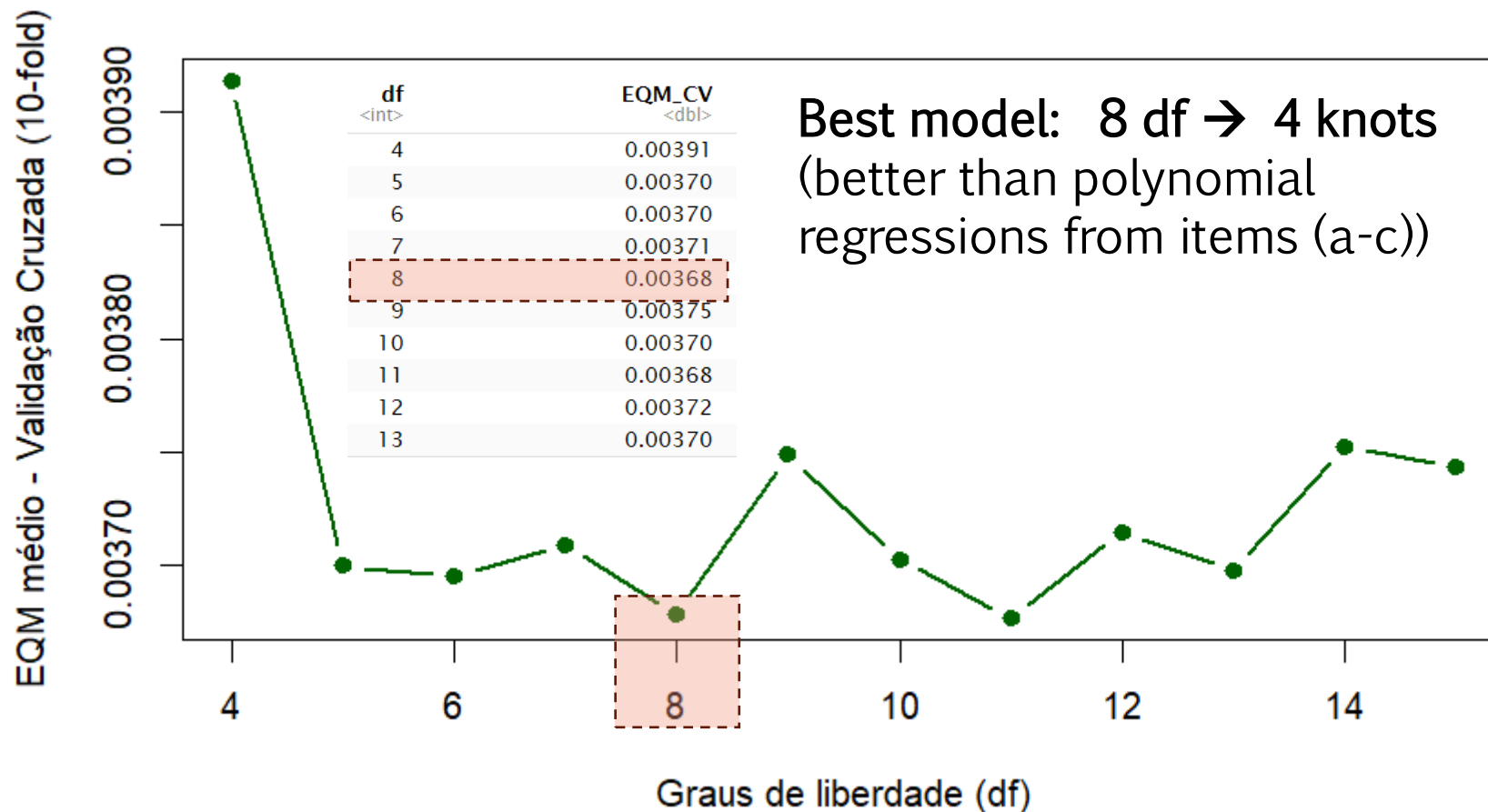
EQM por Grau de Liberdade - Splines





Ex.9f) Cubic Splines: df = 4 to 15

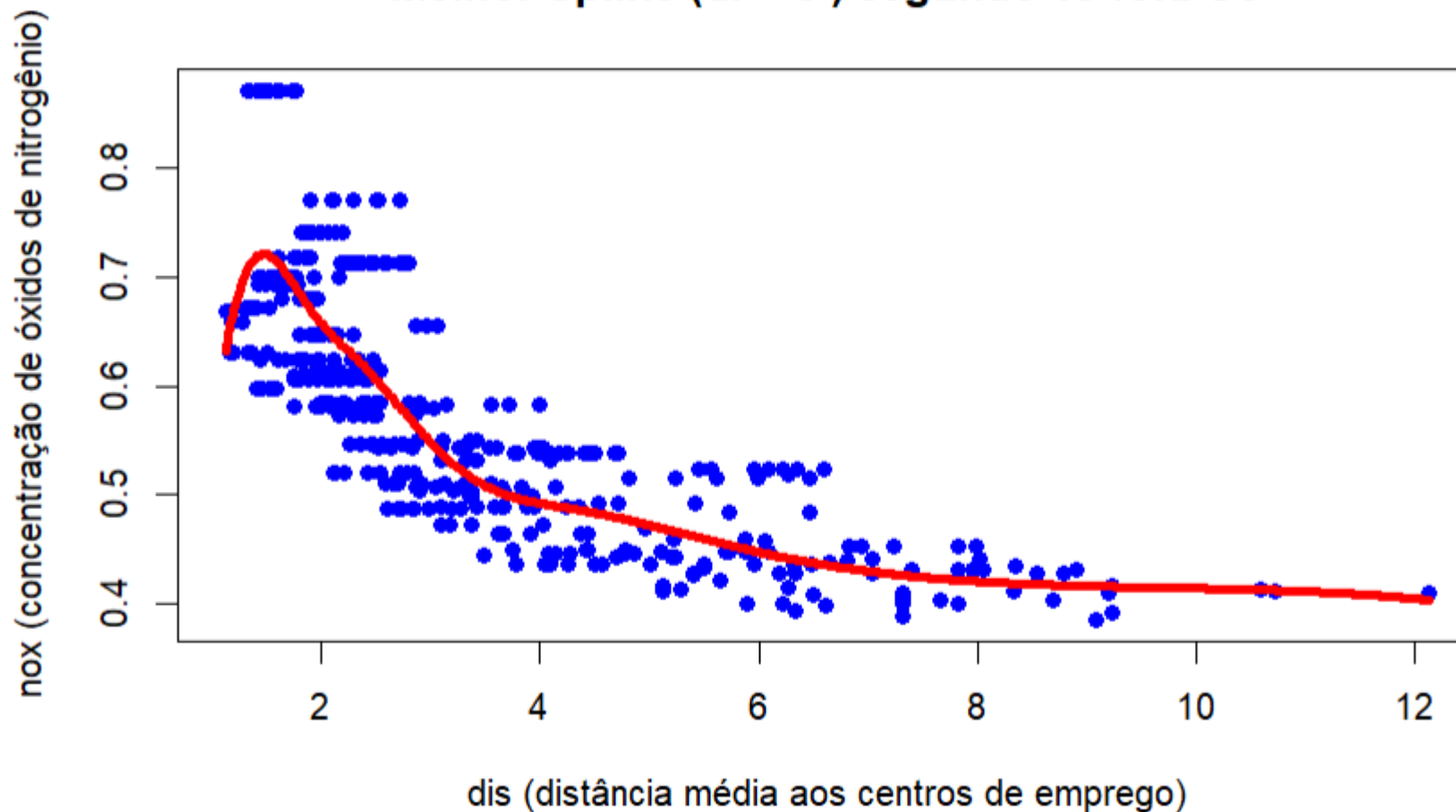
Validação Cruzada (k-fold) - Seleção do Grau de Liberdade (Spline)



Ex.9e) Cubic Splines: df = 4 to 15



Melhor Spline (df = 8) segundo 10-fold CV



Ex.9) Takeaways



- Higher degree polynomials or higher degree of freedom (more cubic splines) tend to improve the fitting of the data
- However, when performing validation, their performance tend to deteriorate due to overfitting
- As a result, the best models tend to lower degrees, balancing their performance in-the-sample and out-of-sample